

BEERAPPA METHRE

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Backend Generative AI Engineer with 2+ years of experience building scalable APIs, AI-powered systems, and high-performance backend services using Python, FastAPI, PostgreSQL, and modern cloud infrastructure. Experienced in LLM integrations, Retrieval-Augmented Generation (RAG), authentication systems (JWT/OAuth2/RBAC), async backend architectures, and production-grade testing. Proven track record of leading engineering initiatives, building secure APIs, and delivering reliable software in fast-paced startup environments.

PROFESSIONAL EXPERIENCE

Senior Gen AI Developer | Simplify AI

February 2026 - May 2026

- * Lead development of full-stack product features across backend APIs and frontend interfaces, building scalable web applications using Python, FastAPI, JavaScript, and React, while collaborating closely with product and AI teams to rapidly ship production features.
- * Design and maintain high-performance RESTful APIs and backend services, implementing efficient data models, database query optimization, and caching strategies using PostgreSQL and Redis to support large-scale application workloads.
- * Build and integrate AI-powered platform capabilities through LLM APIs, enabling intelligent automation and enhanced user workflows while ensuring secure API design, data protection, and compliance across backend systems
- * Improve platform reliability, observability, and performance by implementing structured logging, debugging workflows, monitoring, automated testing (unit, integration, E2E), and CI/CD pipelines with Docker-based deployments on cloud infrastructure.

Software Engineer | Cogneo Technologies

August 2024 – December 2025

- Designed and developed high-performance asynchronous REST APIs using **FastAPI and Python**, reducing API response latency by **40%** and enabling scalable microservice architectures.
- Built secure **authentication and authorization** systems using **JWT, OAuth2, RBAC, OTP flows**, and **multi-tenant access control** for enterprise-grade applications.
- Implemented comprehensive **user management** features including role assignment, permission enforcement, token lifecycle management, pagination, and **rate limiting**.
- Designed and optimized **PostgreSQL** database schemas and complex queries, significantly improving read/write performance for large-scale datasets.
- Built and maintained **data pipelines** using **PySpark and Delta Lake**, implementing **SCD Type 2**, watermarking, lineage tracking, and incremental processing.
- Developed **47+ reusable Data Quality (DQ) checks** covering schema validation, null handling, referential integrity, pattern matching, and business rule enforcement.
- Implemented a robust **testing strategy** with unit, integration, and end-to-end tests using **pytest**, achieving **90%** code coverage and reducing production regressions.
- Integrated **AWS services** (SES for email notifications, S3 for storage), containerized services using **Docker**, and deployed via **CI/CD pipelines** in a production environment.

EDUCATION

Indian Institute of Information Technology, Guwahati

December 2020 – June 2024

B.Tech - Computer Science and Engineering

Guwahati, Assam

PROJECT

Academic Gateway [🔗](#) | [IDE - VS Code](#)

- * Built a **full-stack financial management system** handling college fees, hostel rent, mess dues, and library fines.
- * Implemented **secure authentication, role-based access control**, and real-time fine tracking dashboards.
- * Reduced manual operations by **40%** through automation and clean API-driven workflows..

GharDekho [🔗](#) | [IDE - VS Code](#)

- * Developed a **scalable rental platform** using React, Node.js, Firebase, and MongoDB.
- * Integrated **Google OAuth 2.0 authentication** and **Firebase-powered real-time chat**, ensuring secure user sessions and instant communication to streamline the rental process.

SKILLS

Languages: C/C++, Python, SQL, JavaScript.

Technologies/Frameworks: HTML, CSS, Reactjs, Firebase, Mongodb, FastAPI, PostgreSQL, Pydantic, MYSQL, Git, AWS(EC2, S3, SES), Redis, CI/CD (GitHub Actions), Pyspark, GenAI, LLMs

Developer Tools: VS Code, PyCharm, Jupiter Notebook, Google Colab, Postman, Linux, Docker, Databricks, Microsoft Fabric

CODING PLATFORMS AND CERTIFICATIONS

- Solved **550+** problems on [LeetCode](#) [🔗](#), [GeeksforGeeks](#) [🔗](#), and [HackerRank](#) [🔗](#)